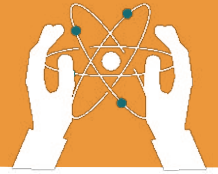


UNIT 9

SPACE AND SATELLITES WITH AI

Learning objectives:

- Distinguish between natural celestial bodies and artificial satellites with examples.
- Identify key applications of artificial satellites like GPS, communication, and Earth observation.
- The role of artificial intelligence (AI) in satellite advancements.
- Explore advancements in satellite technology, including space telescopes.



Introduction to Space and Satellites:

Space is the vast, empty region beyond Earth's atmosphere, where planets, stars, and galaxies exist. There is no air, gravity is weaker, and temperatures can be extremely hot or cold. Scientists explore space to learn about our universe, understand Earth's climate, and find possibilities for life beyond our planet.

Fun Facts about Space:

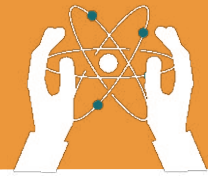
- The Sun is a star that provides light and heat to Earth.
- The Moon is Earth's only natural satellite.
- There is no sound in space because there is no air for sound waves to travel.
- Astronauts float in space due to low gravity.

This chapter explores the universe beyond Earth, covering celestial bodies like stars, planets, moons, and satellites. It highlights the differences between natural and artificial satellites and their applications.



What are Celestial Bodies?

Celestial bodies are naturally occurring objects in the universe such as stars, planets, asteroids, and comets that exist within galaxies. These bodies can be major (like planets) or minor (like comets and asteroids).



SPACE AND SATELLITES WITH A.I

Natural Satellites

Natural satellites, or moons, are celestial bodies that orbit a planet or another larger celestial body.

Example: The Moon is Earth's only natural satellite and is critical for various Earth phenomena like tides.

Artificial Satellites

These are human-made objects designed to orbit Earth or other celestial bodies. They serve numerous practical functions ranging from telecommunications to weather forecasting.

Applications of Artificial Satellites:

Types of Satellites with AI incorporation:

Satellites are designed for different purposes, and Artificial Intelligence (AI) is making them smarter by improving their efficiency, decision-making, and automation. Here are the main types of satellites and how AI is enhancing them:

1. Communication Satellites:

- They transmit TV signals, internet, radio, and phone calls across the world.
- They improve global connectivity.



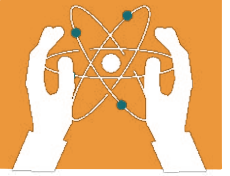
AI Role:

- **Better Signal Processing:** AI helps improve the quality of phone calls and internet speeds.
- **Automatic Frequency Adjustment:** AI prevents interference between satellite signals.

• DID YOU KNOW?

First Human-Made Object in Space: Sputnik 1, launched by the Soviet Union in 1957, was the first artificial object to orbit the Earth.



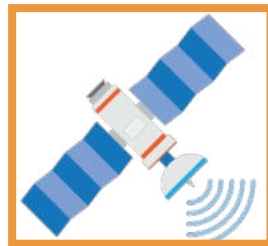


Example:

AI-powered satellites provide fast internet to rural areas (e.g., Starlink by SpaceX).

2. Weather Satellites

They monitor weather patterns, storms, and climate changes. They predict natural disasters like hurricanes, cyclones, and floods.



AI Role:

- **Faster Weather Predictions:** AI analyzes cloud movements and atmospheric conditions for accurate forecasts.
- **Disaster Warnings:** AI can detect and alert areas about approaching storms or earthquakes.
- **Climate Monitoring:** AI helps track global temperature changes and environmental damage.

Example:

- NASA's GOES-R weather satellites use AI for storm tracking.

3. Navigation Satellites (GPS)

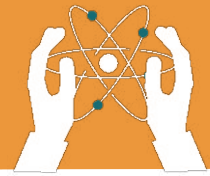
- They help with location tracking (Google Maps, car navigation, airplane routes).
- They assist in defence and military operations.

AI Role:

- **Real-Time Route Optimization:** AI improves GPS accuracy for cars, planes, and ships.
- **Traffic Prediction:** AI helps predict and reduce road congestion.
- **Autonomous Vehicles:** AI-powered GPS assists self-driving cars in navigation.

Example:

- AI-enhanced GPS helps drones and autonomous cars move safely.



SPACE AND SATELLITES WITH A.I

Earth Observation Satellite:

- It enables to study Earth's surface, forests, oceans, and pollution levels.
- It helps with agriculture, water management, and disaster relief.

AI Role:

- Detecting Deforestation & Wildfires: AI identifies tree loss and fire outbreaks.
- Smart Agriculture: AI monitors soil health and crop conditions.
- Environmental Protection: AI tracks pollution and ocean health.

Example:

- AI helps predict crop yields and detect illegal logging.

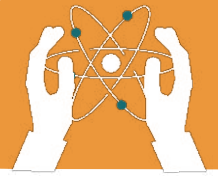


● DID YOU KNOW?

Space agencies like NASA (USA), ESA (Europe), and Roscosmos (Russia) play significant roles in advancing satellite technology and space exploration.

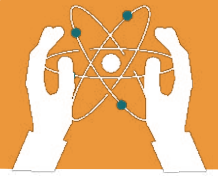


SPACE AND SATELLITES WITH A.I



Fill in the blanks.

1. The Moon is Earth's only _____ satellite.
2. The _____ of a satellite is essential for tasks such as GPS and weather forecasting.
3. Satellites help in studying _____ phenomena like earthquakes and storms.



Choose the correct answer.

1. What are celestial bodies?

- Man-made machines used in space
 - Objects used in weather forecasting
 - Naturally occurring objects like stars and planets
 - Satellites that orbit the Earth
-

2. Which of the following is an example of a natural satellite?

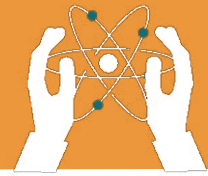
- Mars Rover
 - The Moon
 - Hubble Space Telescope
 - GPS Satellite
-

3. What is the primary purpose of an Earth Observation Satellite?

- To study other planets
 - To monitor space weather
 - To study Earth's surface, forests, and oceans
 - To explore the Moon
-

5. Which of the following is NOT a function of artificial satellites?

- Weather forecasting
 - Soil health monitoring
 - Orbiting naturally without human design
 - Telecommunications
-



Answer the Following Questions:

Short-Answer Questions

- Q1. What are celestial bodies? Give two examples.
- Q2. Differentiate between artificial and natural satellite.
- Q3. What is the main function of weather satellites?
- Q4. What is the role of AI in communication satellite?
- Q5. What is space telescope? Provide an example.

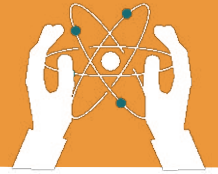
Long-Answer Questions

- Q1. Explain the difference between natural and artificial satellites, providing examples of each.
- Q2. Discuss the impact of the Global Positioning System (GPS) on navigation and daily life.



Critical Thinking

How do artificial satellites benefit our daily lives, and what might change if we didn't have them?



Evaluate each statement as true or false. If true, explain why. If false, provide the correct reasoning.

Statement: " Satellites are important for environmental monitoring?"

Ans: _____

Statement: " The Moon is considered as natural satellite, and it influences Earth."

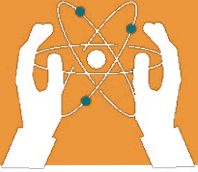
Ans: _____

Statement: "Artificial satellites are essential for telecommunications across the globe."

Ans: _____

Statement: " A geostationary satellite remains fixed over one point on Earth's surface, and it is advantageous for communication."

Ans: _____



SPACE AND SATELLITES WITH A.I



Build Your Own Model Satellite



ACTIVITY

Build Your Own Model Satellite: Using materials such as cardboard, wire, and solar panel stickers, students construct a simple model of an artificial satellite.

Satellite Tracking Exercise: Students use online tools to track the real-time positions of different satellites orbiting Earth.

● DID YOU KNOW?

Largest Satellite

Constellation: The Star link project by SpaceX aims to deploy thousands of smallsatellites to provide global broadband internet coverage.

